Exp.no : 12 **Z-TEST**

AIM:

A manufacturer claims that the average weight of packets is 50 g.A random sample of 36 packets has
an average weight of 51.2 g with a known σ = 3 g.At a 5% significance level, test the claim.

PYTHON CODE:

```
import numpy as np
from math import sqrt
from scipy.stats import norm
# Given data
x_bar = 51.2 # sample mean
mu_0 = 50 # population mean
sigma = 3 # population standard deviation
n = 36 # sample size
# Calculate Z-statistic
z_stat = (x_bar - mu_0) / (sigma / sqrt(n))
# Two-tailed p-value
p_value = 2 * (1 - norm.cdf(abs(z_stat)))
print(f"Z-statistic: {z_stat:.3f}")
print(f"P-value: {p_value:.4f}")
alpha = 0.05
if p_value < alpha:
print("Reject Null Hypothesis → Mean is significantly different from 50 g.")
else:
print("Fail to Reject Null Hypothesis → No significant difference.")
```

OUTPUT:

Z-statistic: 2.400
P-value: 0.0164
Reject Null Hypothesis → Mean is significantly different from 50 g.

RESULT:

The code is executed successfully.